

OISD-STD-174

Well Control Standards

Industry: Oil & Natural Gas

Category: Drilling & Well Operations

Edition/Revision: OISD-RP-174 2nd Edition (2008) / OISD-STD-174

Research Date: August 26, 2026

DISCLAIMER: This is an AI-generated research summary based on publicly available information. It is NOT an official OISD publication and must not be considered a replacement for the original OISD standard. The official standard should always be consulted for authoritative requirements.

1. STANDARD OVERVIEW

Standard Number: OISD-STD-174 (formerly OISD-RP-174)

Official Title: Well Control Standards

Type: Standard (STD) / Recommended Practice (RP)

Industry Category: Drilling & Well Operations / Exploration & Production

Main Subject: Selection, installation, testing, and maintenance of well control equipment, along with procedures for well kick prevention, blowout control, and personnel training in the Indian oil and gas sector.

2. PURPOSE OF THE STANDARD

During drilling and workover operations, the flow of formation fluids into the wellbore is known as a "kick." If not rapidly detected and controlled, a kick can escalate into a "blowout," which is an uncontrolled release of highly flammable well fluids and gases.

The primary purpose of **OISD-STD-174** is to ensure the safe handling of such threatened kicks by providing standardized procedures and equipment specifications. It aims to protect personnel, equipment, and the environment by establishing rigorous requirements for well control systems, blowout preventers (BOPs), well planning, and emergency response drills.

3. SCOPE

The standard covers a wide array of activities and equipment essential to well control, including:

- **Applicability:** Rigs engaged in exploration and production of hydrocarbons (both onshore and offshore).

- **Equipment:** Selection, installation, testing, and maintenance of both surface and subsea well control equipment (Blowout Preventers, Diverters, Accumulators, Choke and Kill manifolds).
- **Operations:** Drilling operations, workover operations, and underbalanced drilling.
- **Personnel:** Competence requirements, well control training, and mandatory well control drills (e.g., pit drills, trip drills).
- **Exclusions:** While it covers well control, it is a self-regulatory guideline that supplements (but does not supersede) statutory requirements from regulatory bodies like the Directorate General of Mines Safety (DGMS).

4. INDUSTRY APPLICATION

OISD-STD-174 is directly applicable to the following domains within the Oil & Gas industry:

- **Exploration & Drilling:** Ensuring safe penetration of over-pressurized zones without uncontrolled fluid release.
- **Well Operations & Workover:** Safe execution of well interventions, clean-outs, or repairs where formation pressure is a hazard.
- **Offshore Operations:** Specifications for subsea BOP equipment and blowout prevention mechanisms.
- **HPHT Environments:** Special arrangements and rigorous control mechanisms for High-Pressure High-Temperature wells.

5. MAJOR AREAS COVERED

Based on publicly available documentation, the standard comprehensively covers the following major operational and safety aspects:

- **Safety Management & Competence:** Drills, well control training, and personnel readiness.
- **Process & Equipment Safety:** Diverter systems, Blowout Preventer (BOP) stacks, choke & kill lines.

- **Operational Controls:** Methods for well control, soft shut-in vs. hard shut-in procedures, kick indication and prevention.
- **Inspection, Testing & Maintenance:** Low-pressure and high-pressure BOP testing, periodic inspection of wellhead components.
- **Monitoring:** Mud gas separators, trip tank systems, degassers, and downhole pressure monitoring.

6. DETAILED TOPIC-WISE SUMMARY

Causes, Detection, and Control of Kicks

What it covers: Explains how a "kick" occurs when the hydrostatic pressure of the drilling mud column falls below the formation pore pressure, allowing fluids to enter the wellbore.

Main Objective: To equip crews with the knowledge to recognize early warning signs (e.g., increased return flow, pit volume gain, drilling breaks) and take immediate action.

Blowout Preventer (BOP) Systems

What it covers: The selection and stacking of preventers (Annular, Ram, Blind Shear) based on the well's anticipated pressure.

Main Objective: To provide a physical, secondary barrier to seal the wellbore. The standard mandates specific arrangements, side outlets, and flanged connections for safely routing fluids during a well-killing operation.

Shut-in Procedures

What it covers: Prescribed steps for "Soft Shut-in" and "Hard Shut-in" during drilling and tripping.

Important Requirements: In a hard shut-in, the BOP (preferably annular) is closed immediately on kick detection. The pressure is allowed to stabilize, and Shut-In Drill Pipe Pressure (SIDPP), Shut-In Casing Pressure (SICP), and Pit Gain are recorded.

Well Control Drills

What it covers: Mandatory field exercises such as the *Pit Drill (On Bottom)* and *Trip Drill* (when string is in/out of hole).

Main Objective: Ensure that drilling crews react instinctively and rapidly to secure the well upon kick detection.

7. EQUIPMENT AND SYSTEMS

Equipment / System	Purpose	Standard Requirement (Publicly Known)
Blowout Preventer (BOP)	Seals the wellbore to confine well fluids.	Must include adequate combinations of annular, pipe, and blind shear rams based on well rating.
Accumulator (BOP Control Unit)	Stores hydraulic fluid under pressure (using nitrogen/inert gas) to operate BOPs/diverters.	Must provide sufficient hydraulic energy to operate the BOP stack quickly under emergency conditions.
Choke and Kill Manifold	Circulates out a kick safely under controlled pressure.	Requires HCR (Hydraulic Control Valves) or manual valves, securely installed and tested.
Trip Tank	Monitors precise fluid volumes entering or leaving the well during tripping operations.	Must be strictly isolated and monitored to track exact displacement volumes, avoiding unseen kicks (Clause 6.8).
Mud Gas Separator (MGS)	Separates highly volatile gas from the drilling fluid during a kick circulation.	Must be sized correctly to vent gas safely away from the rig floor.

8. SAFETY REQUIREMENTS

- **Process Safety:** Managing the fundamental balance between Formation Pressure (P_{pore}) and Mud Hydrostatic Pressure (P_{hyd}).

- **Pressure Safety:** The standard mandates the calculation and utilization of exact Hydrostatic and Bottom Hole Pressures (BHP) to prevent wellbore fracturing and blowouts.
- **Explosion Prevention:** Use of degassers and mud gas separators to prevent explosive gas accumulations at the surface rig floor. Proper electrical compliance (e.g., FLP/Ex-d fittings in hazardous zones) is also referenced in association with wellhead proximity.
- **Mechanical Safety:** Fall Protection Devices (FPD) and mechanical integrity of casing running tools associated with operations near the well center.

9. OPERATIONAL REQUIREMENTS

Operating Practices: Precise calculation of pressures is foundational to operations under OISD-STD-174. Hydrostatic pressure must be continuously evaluated.

Formula Reference (Publicly documented):

$$P_{hyd} \text{ (psi)} = 0.052 \times \text{Mud Weight (ppg)} \times \text{True Vertical Depth (feet)}$$

$$P_{hyd} \text{ (kg/cm}^2\text{)} = \text{Mud Weight (gm/cc)} \times \text{True Vertical Depth (mtrs)} / 10$$

Monitoring: Constant monitoring of pit volumes and return flow rates. Any anomaly must trigger immediate well control protocols.

Workover Precautions: Workover fluids must be properly weighted to account for depleted or abnormally pressured zones based on actual offset well data and current conditions.

10. INSPECTION, TESTING & MAINTENANCE

Pressure Testing: Routine testing of the BOP stack and manifold is mandated.

- *Low-Pressure Test:* Conducted first (typically around 200–350 psi) to verify initial seal integrity of rams, annular preventers, and manifolds.
- *High-Pressure Test:* Follows the low-pressure test to ensure the equipment can handle the maximum expected wellbore pressure up to its rated capacity.

Maintenance: Periodic inspections of elastomeric seals, control lines, and accumulator charge pressures are standard operational directives to avoid failure during an actual blowout scenario.

11. EMERGENCY & HSE ASPECTS

Emergency preparedness is enforced through rigorous crew competence standards. A major component of HSE is the "Drills and Training" section, which ensures that upon a kick indication, the crew immediately moves to shut-in the well without waiting for supervisor instruction. This minimizes the volume of the kick and greatly reduces the risk of a blowout, protecting both the workers and the surrounding environment from catastrophic hydrocarbon release.

12. IMPORTANT TECHNICAL CONCEPTS

Hydrostatic vs. Formation Pressure: Safe drilling relies on the hydrostatic pressure of the mud column being slightly higher than the pore pressure (overbalance). If it drops below (underbalance), a kick occurs.

Bottom Hole Pressure (BHP): The sum of all pressures at the bottom of the well. While drilling, $BHP = P_{hyd} + P_{annular_friction}$. During a shut-in, $BHP = P_{hyd} + \text{Surface Pressure}$.

Fracture Pressure: The pressure at which the formation rock breaks. During a kick circulation, keeping the BHP above pore pressure but below fracture pressure is critical to prevent underground blowouts.

13. IMPORTANT TERMS & ABBREVIATIONS

Term / Abbreviation	Meaning	Relevance
BOP	Blowout Preventer	Primary safety barrier at the wellhead.
BHP	Bottom Hole Pressure	Critical metric for balancing the well.
SIDPP / SICP	Shut-In Drill Pipe / Casing Pressure	Pressures read at the surface after a shut-in, used to calculate required kill mud weight.
HCR	Hydraulic Control Valve	Remotely operated valve on the choke line.
HPHT	High-Pressure High-Temperature	Extreme well conditions requiring advanced well control equipment.

14. RELATED STANDARDS & REGULATIONS

- **API Spec 7K:** Standard for Drilling and Well Servicing Equipment (referenced for rotary tables, etc.).
- **API RP 53 / 54:** Standard practices for blowout prevention equipment and occupational safety for drilling.
- **DGMS Regulations:** Directorate General of Mines Safety requirements which supersede or complement OISD standards in India.

15. PRACTICAL USE CASES

Scenario 1: Kick Detection during Drilling

Relevant aspect: Monitoring Instrumentation & Shut-in procedures.

Why it matters: If the mud logger notices a sudden pit gain, the driller must immediately execute a hard or soft shut-in as per OISD-STD-174 to prevent a blowout.

Scenario 2: Workover on a Depleted Well

Relevant aspect: Trip Tank operation (Clause 6.8) and Workover Precautions.

Why it matters: Pulling tubing out of the well reduces hydrostatic head. The trip tank must accurately measure fill-up fluid to ensure the well is not swabbing (pulling formation fluid in).

Scenario 3: BOP Certification for New Rig Commissioning

Relevant aspect: Equipment Selection and Testing.

Why it matters: Before spudding, the rig contractor must demonstrate low and high-pressure testing of the BOP stack and Choke manifold to satisfy OISD and DGMS regulatory requirements.

16. KEY TAKEAWAYS

1. OISD-STD-174 provides the foundational safety framework for Well Control in India.
2. Applies to both drilling and workover operations, onshore and offshore.
3. Dictates the mandatory use, selection, and configuration of Blowout Preventers.
4. Provides specific calculation models for Hydrostatic and Bottom Hole Pressures.
5. Outlines the procedural difference between Soft Shut-in and Hard Shut-in.
6. Requires routine well control drills (Pit drill, Trip drill) to ensure personnel competence.
7. Mandates a two-step pressure testing regimen (low and high) for wellhead barriers.
8. Highlights the importance of the trip tank during tripping and perforation operations.
9. Covers specialized situations like Underbalanced Drilling and HPHT wells.
10. Works in conjunction with statutory regulations like DGMS.

17. INFORMATION LIMITATIONS

- **Verified Information:** General scope, definitions, key mathematical formulas (hydrostatic pressure), mandatory equipment types (BOP, accumulator, choke manifold), and operational shut-in sequences.

- **Partially Verified Information:** Specific testing pressure limits (e.g., 200-350 psi low-pressure testing is cited in rig SOPs derived from OISD-174, but exact numeric thresholds inside the latest edition may vary).
- **Information Requiring Official Access:** Exhaustive engineering schematics for diverter sizing, exact required capacities for accumulators, detailed metallurgical requirements, and the full text of all annexures.

Note: "Detailed requirement is not publicly available in the sources reviewed." Do not guess missing numerical parameters for BOP shear rams or accumulator volumes.

18. SOURCES & REFERENCES

1. **Oil Industry Safety Directorate (OISD)** - Accident Case Studies (Reference to Clause 6.8 of OISD-STD-174). [1.2.2]
2. **Oil India Limited** - Rig Tender Documents (References to OISD-STD-174 compliance for BOP, Choke Manifolds, and Testing). URL: oil-india.com [1.2.1, 2.2.2, 2.2.3]
3. **Scribd Document Repository** - OISD-RP-174 / OISD-STD-174 Well Control Standards (Extracts, Introductions, Scopes, and Definitions). [1.1.1, 1.1.2, 1.1.3, 2.1.4, 2.1.5]
4. **Drilling Operations SOP Manuals** - Citations of OISD-174 testing procedures for BOP operations. [1.3.1, 1.3.2]